Immunization Overview

Dr. Moustafa Al-Shammari Ph.D.

Pediatric Nursing

Immunization Overview

Immunization is the process of preparing the body to fight specific diseases through the induction of immunity.

There are **two types of immunization**:

Passive Immunization (Natural and Artificial)	**Active Immunization** (Natural and Acquired)
Natural: Transfer of antibodies from mother to child through the placenta. Artificial: Administration of immunoglobulins containing antibodies, often for emergency or contact cases.	Natural: Acquiring immunity through infections. Acquired: Induced through vaccines.

Vaccines

- **Vaccine**: A suspension of live or killed microorganisms or their antigenic portions, used to induce immunity to specific diseases.
- **Toxoid**: A modified bacterial toxin used to induce immunity against toxins (e.g., tetanus, diphtheria).

Types of Vaccines:

- **Live Attenuated**: Contains weakened live viruses (e.g., Sabin Polio, Measles, Mumps).
- **Inactivated or Killed**: Contains dead viruses or bacteria (e.g., Salk Polio, Rabies).
- **Cellular Fraction**: Contains fragments of bacteria (e.g., Pneumococcal, Meningococcal).

Types of Vaccines:

- **Recombinant**: Contain a genetically engineered version of the virus (e.g., Hepatitis B).
- **Toxoids**: Modified toxins (e.g., Tetanus, Diphtheria).
- **Immunoglobulin**: Contains antibodies (e.g., Hepatoglobulin).
- **Antitoxins**: Contains antibodies to neutralize toxins (e.g., Tetaglobulin).

The Iraqi Immunization Schedule (2018)

At birth: Hepatitis B (HVB1), BCG (Tuberculosis), OPV (Oral Polio Vaccine).
2 months: Hexa (DTaP, Hib, HBV, IPV), OPV, Rota, PCV.
4 months: Hexa (DTaP, Hib, HBV, IPV), OPV, Rota, PCV.
6 months: Hexa (DTaP, Hib, HBV, IPV), OPV, PCV.
9 months: Measles, Vitamin A.
15 months: MMR (Measles, Mumps, Rubella).
18-24 months: First booster (Polio, Penta, Vitamin A).
4-6 years: Second booster (Polio, Tetra, MMR).

Abbreviations in Immunization

HexaVPI ,VBH ,biH ,PaTD :

PentaVPI ,biH ,PaTD :

TetraVPI ,PaTD :

Rotaeniccav surivatoR :

PCVeniccav etagujnoc laccocomuenP :

BCG)sisolucrebuT(nireuG-ettemlaC ellicaB :

BCG Vaccine

Route: Intradermal

Dose: 0.05ml

Side Effects: Regional lymphadenitis, abscess, osteomyelitis.

Efficacy: 50% for pulmonary TB, 50-80% for disseminated TB.

Contraindications: Subcutaneous administration leads to abscess formation.

Polio Vaccines - Types

- **Sabin (Oral) Vaccine**:
  - Live attenuated virus.
  - Mass immunization.
  - Easier to administer.
  - Potential community-wide immunity.

Polio Vaccines - Types

- **Salk (Inactivated) Vaccine**:
- Injected.
- Longer lifespan than Sabin.
- Effective in preventing polio but less suited for mass vaccination.

DPT Vaccine and Side Effects

- **DPT**: Diphtheria, Pertussis, Tetanus.
- **Side Effects**:
  - Mild: Local swelling, slight fever.
  - Moderate: Shock.
  - Severe: Convulsions, encephalopathy (rare).
- **Note**: If severe reaction occurs, use D.T vaccine instead of DPT.

MMR Vaccine and Side Effects

- **Live Attenuated** (Measles, Mumps, Rubella).
- **Side Effects**:
  - Fever and rash (5-15%).
  - Transient arthritis (rare).
  - Encephalopathy (very rare).
- **Contraindications**: Pregnancy, immunosuppression, severe allergic reactions.

Hepatitis B Vaccine and Side Effects

- **Common Side Effects**:
  - Erythema, induration at injection site.
  - Fever, malaise, flu-like symptoms.
  - Rare: Arthritis, myalgia.

Rota Virus Vaccine

- **Side Effects**:
- Diarrhea, vomiting, nasopharyngitis, otitis media.

- **Contraindications**: Immunocompromised children should not receive live vaccines due to the risk of adverse reactions from the live virus.

Immunization for Immunocompromised Children

- **Contraindicated Vaccines**: Live bacterial or viral vaccines (e.g., MMR, rotavirus).
- **Postponing Vaccination**: Wait 3 months after immunosuppressive treatment.

Pre-Vaccination Assessment

- **Questions to Ask**:
  - Is the child sick today?
  - Any severe allergies or past reactions to vaccines?
  - Weakened immune system?
  - Recent blood transfusion?
  - Developmental concerns?
- **Important**: Minor illnesses like a cold or mild diarrhea are not contraindications.

Nursing Actions Before Vaccine Administration

- **General Instructions**:
  - Ensure the child is healthy and doesn't have contraindications.
  - Provide clear explanations to parents.
  - Administer the vaccine following correct procedures.

Potential Reactions and Nursing Interventions

- **Pain with Injection**:
  - Use topical anesthetics and distractions (e.g., bubbles, music).
- **Fever/Myalgia**:
  - Administer acetaminophen.
- **Severe Reactions**:
  - Monitor for allergic reactions, seizures, and provide emergency care as necessary.

Common Questions Parents May Ask

- **What are immunizations?** Vaccines stimulate the body to defend against infections.
- **Are immunizations safe?** Most vaccines are safe; side effects are generally mild and temporary.



Thank you